

Supply Chain Decision Intelligence

Preventive delivery-risk control, profitability exposure, and governed operational prioritization.

EXECUTIVE STATUS

A01 + A02 + A03

GOVERNED DATA VERIFIED

SCORED ORDER ITEMS

34,467

HIGH DELIVERY RISK

40.0%

EXPECTED PROFIT

\$740,319

EXPECTED MARGIN

10.4%

ACTIVE REVIEW QUEUE

13,804

PROTECTED VALUE AT RISK

\$2,816,571

DECISION INTELLIGENCE FLOW

Frozen model outputs become a governed operating queue.

A01 · DELIVERY RISK

13,804

high-risk items · 40.0% exposure

Approved threshold 35.0%



A02 · PROFITABILITY

\$740,319

expected profit · 10.4% margin

112 negative-profit items



A03 · PRIORITIZATION

13,804

items in active review · 40.0%

Protect first, expedite selectively

EXECUTIVE FOCUS

Where operating attention should concentrate first.

1 · PROTECT VALUE AT RISK

\$2,816,571

39.4% of governed order value

2 · REVIEW SELECTIVE EXPEDITE

52 items

High risk with limited predicted margin

3 · PRESERVE SERVICE QUALITY

20,603 items

Avoid unnecessary intervention cost

A03 PRIORITY PORTFOLIO

Current operational mix under the governed risk-margin policy.



PROTECT HIGH VALUE AT RISK

13,752 · 39.9%

EXPEDITE SELECTIVELY

52 · 0.2%

PRESERVE SERVICE

20,603 · 59.8%

STANDARD PROCESS

60 · 0.2%

EXECUTIVE TAKEAWAY

13,804 scored order items require active operational review.

The governed A03 policy concentrates attention on **13,752 high-value-at-risk items**, representing **\$2,816,571** in order value. Preserve service quality for the stable portfolio and use expedited capacity selectively.

Where should management act first?

Prioritize delivery-risk protection, preserve margin, and allocate operational attention using governed held-out outputs.

DECISION STATUS

Governed action
view

DATA QA VERIFIED

ACTIVE REVIEW QUEUE

13,804

40.0% of scored items

VALUE TO PROTECT

\$2,816,571

high-value risk portfolio

EXPECTED PROFIT

\$740,319

10.4% expected margin

SELECTIVE EXPEDITE

52

review before premium cost

NEGATIVE-PROFIT EXPOSURE

\$1,998

modeled exposure, not realized loss

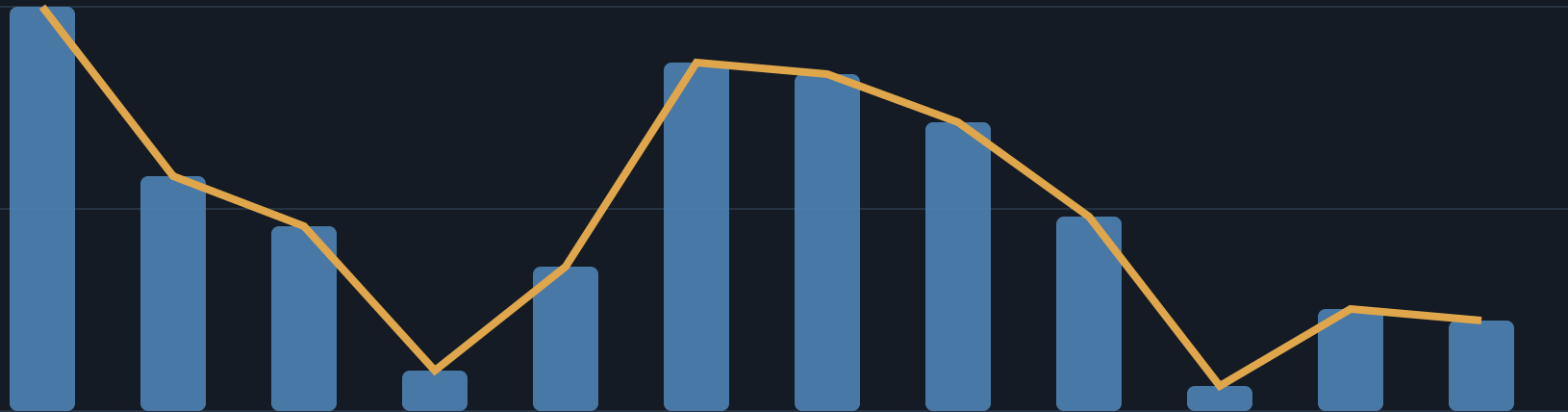
WEEKLY PROTECTED VALUE AT RISK

Order value requiring preventive protection · trailing four-week operational run rate

LATEST WEEK

\$15,567

4W run rate \$15,567



Blue bars: protected order value

Amber line: trailing 4W operational run rate

Run rate supports operational planning. It is not a statistical forecast.

WHERE TO DEPLOY ATTENTION

Use the Geographic Hotspots page to separate workload scale from risk severity.

Scale hotspots

France and Germany concentrate the largest high-risk operating workload.

Severity hotspot

Sweden requires targeted review: approximately 91% of its scored items are high risk despite lower volume.

RECENT OPERATING PRESSURE

Latest four weeks versus the immediately prior four weeks.

PROTECT QUEUE · LATEST 4W

735 items

▲ 2.7% vs prior 4W

PROTECTED VALUE · LATEST 4W

\$90,290

▼ 49.0% vs prior 4W

Queue pressure increased while value intensity declined. Maintain broad preventive controls, but reserve premium-cost intervention for item-level review.

MANAGEMENT ACTION AGENDA

Allocate attention using differentiated operating treatments.

01 · Protect high-value orders before dispatch

13,752 items · \$2,816,571 governed order value

02 · Review selective-expedite cases individually

52 items · approve premium-cost action only when service impact justifies it

03 · Preserve stable service without over-intervention

20,603 items · maintain normal controls and protect margin

Preventive Prioritization Before Dispatch

Surface late-delivery risk before dispatch using the approved AO1 policy.

Profit Band

All

Order Date

22/04/2017

31/01/2018

Priority Segment

All

Risk Classification

All

Reset Filters

OPERATING POLICY

Threshold 35.0%

GOVERNED DATA VERIFIED

SCORED ORDER ITEMS

34,467

HIGH RISK ITEMS

13,804

HIGH RISK RATE

40.0%

AVERAGE LATE RISK

52.4%

APPROVED THRESHOLD

35.0%

OPERATIONAL RISK EXPOSURE

Held-out scored order-item distribution.



HIGH RISK

13,804

40.0% of scored items

BELOW THRESHOLD

20,663

60.0% of scored items

APPROVED DECISION POLICY

35.0% late-delivery probability

RECALL

61.7%

PRECISION

84.7%

ALERT RATE

41.5%

Frozen upstream. Reused without recalculation in Power BI.

VALIDATION CONFUSION MATRIX

TRUE POSITIVE

9,726

FALSE POSITIVE

1,758

FALSE NEGATIVE

6,035

TRUE NEGATIVE

10,124

METHODOLOGY NOTE

Threshold approved on 27,643 validation rows · Exposure: 34,467 held-out scored items · Frozen upstream outputs

EXECUTIVE TAKEAWAY

13,804 order items require preventive attention.

The approved 35.0% policy flags 40.0% of the held-out scored population while maintaining 84.7% validation precision.

Decision support only; realized intervention outcomes are outside scope.

Where Does Preventive Attention Create Value?

Prioritize high-risk order items before dispatch and focus attention where financial exposure is greatest.

Profit Band

All

Order Date

22/04/2017

31/01/2018

Priority Segment

All

Risk Classification

All

Reset Filters

OPERATING POLICY

Threshold 35.0%

GOVERNED DATA VERIFIED

HIGH-RISK ITEMS

13,804

Require preventive attention

HIGH-RISK RATE

40.0%

▲ +0.5% vs prior 4W

HIGH-RISK ORDER VALUE

\$2,825,135

39.5% of governed order value

NEGATIVE-PROFIT HIGH RISK

52

Review first for selective action

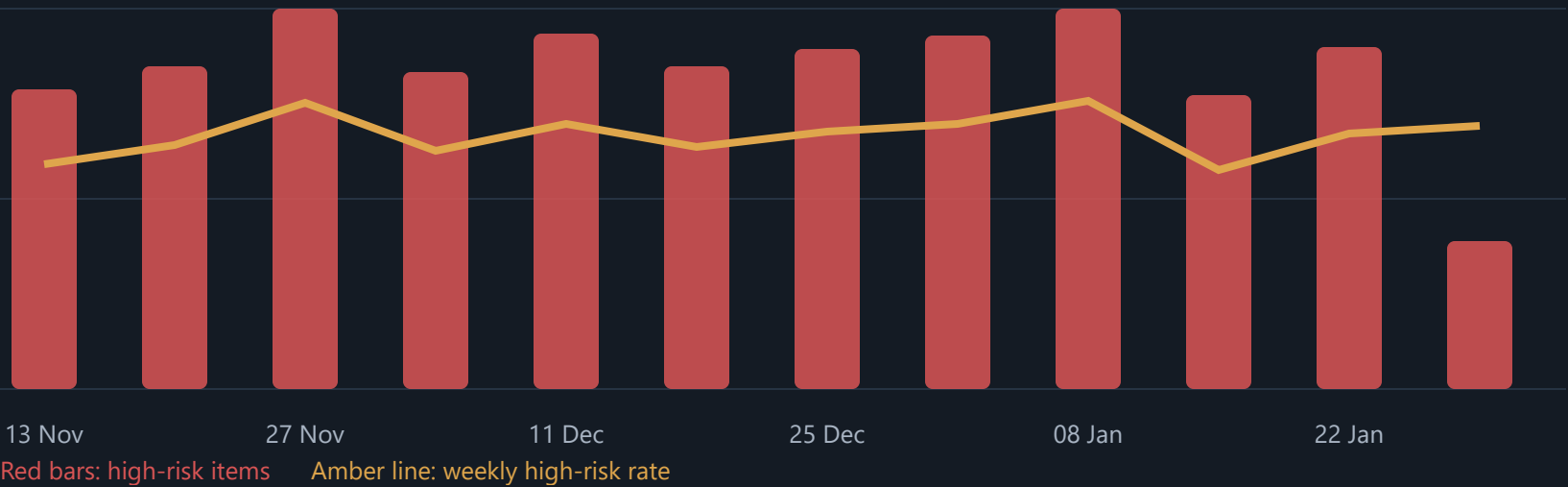
LATEST 4W HIGH RISK

738

▲ +1.9% vs prior 4W

WEEKLY HIGH-RISK EXPOSURE

Preventive-attention queue and risk-rate movement over the latest 12 available weeks.



PREVENTIVE ACTION BRIEF

What supply-chain operations should review before dispatch.

RECENT RISK MOVEMENT

Risk rate increased by 0.5%

Latest 4W 40.1% · Prior 4W 39.5%

VALUE UNDER RISK

\$2,825,135

Prioritize protection capacity

REVIEW FIRST

52 items

High risk and negative profit

Use the governed threshold to focus preventive attention before dispatch. Review the financially exposed queue first, then drill through to Order ID and Item ID.

POLICY EVIDENCE

RECALL

61.7%

PRECISION

84.7%

ALERT RATE

41.5%

Threshold frozen upstream on 27,643 validation rows. Dashboard outputs support prioritization, not realized intervention outcomes.

EXECUTIVE TAKEAWAY

Protect \$2,825,135 in high-risk orders before dispatch.

Risk rate increased by 0.5% versus the prior four weeks. Review the 52 high-risk negative-profit items first.

IMMEDIATE REVIEW QUEUE

High-risk negative-profit items ranked by order value.

TOTAL CRITICAL QUEUE

52 items

ORDER ID	ITEM ID	LATE RISK	ORDER VALUE	EXPECTED PROFIT	PRED. MARGIN
70555	173870	73.8%	\$1,425	-\$80.78	-5.7%
74516	177831	73.8%	\$1,245	-\$6.52	-0.5%
71994	175309	93.7%	\$469	-\$2.40	-0.5%
69665	172980	99.9%	\$438	-\$80.78	-18.4%

Predicted Margin Exposure Review

Surface order items requiring margin attention using frozen AO2 estimates.

Profit Band

All

Order Date

22/04/2017

31/01/2018

Priority Segment

All

Risk Classification

All

Reset Filters

MODEL STATUS

Frozen AO2 outputs

GOVERNED DATA VERIFIED

SCORED ORDER ITEMS

34,467

EXPECTED PROFIT

\$740,319

AVG EXPECTED PROFIT

\$21.48

EXPECTED MARGIN

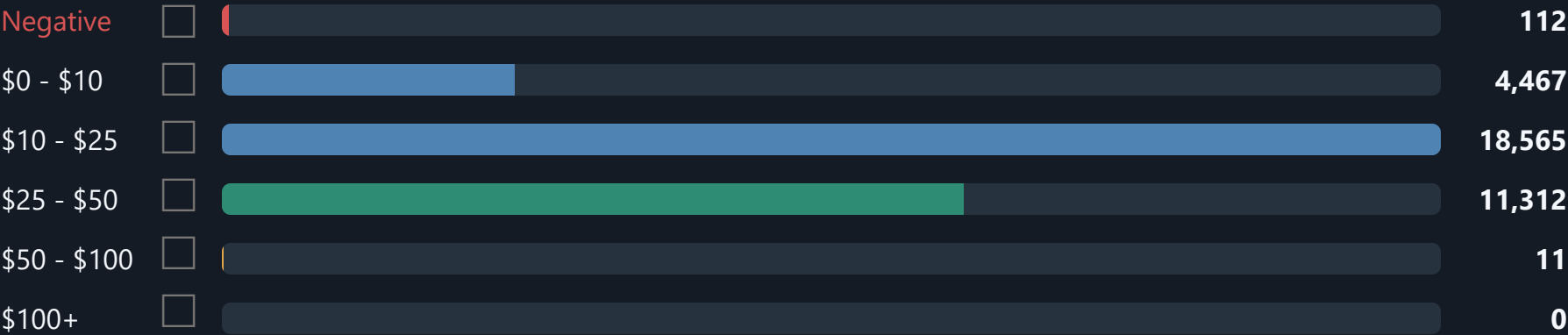
10.4%

NEGATIVE PROFIT ITEMS

112

EXPECTED PROFIT DISTRIBUTION

Scored order items by frozen AO2 profit band.



112 items (0.3%) require margin-protection review.

VALIDATION EVIDENCE

Gradient Boosting · conservative baseline

RMSE

\$95.62

MAE

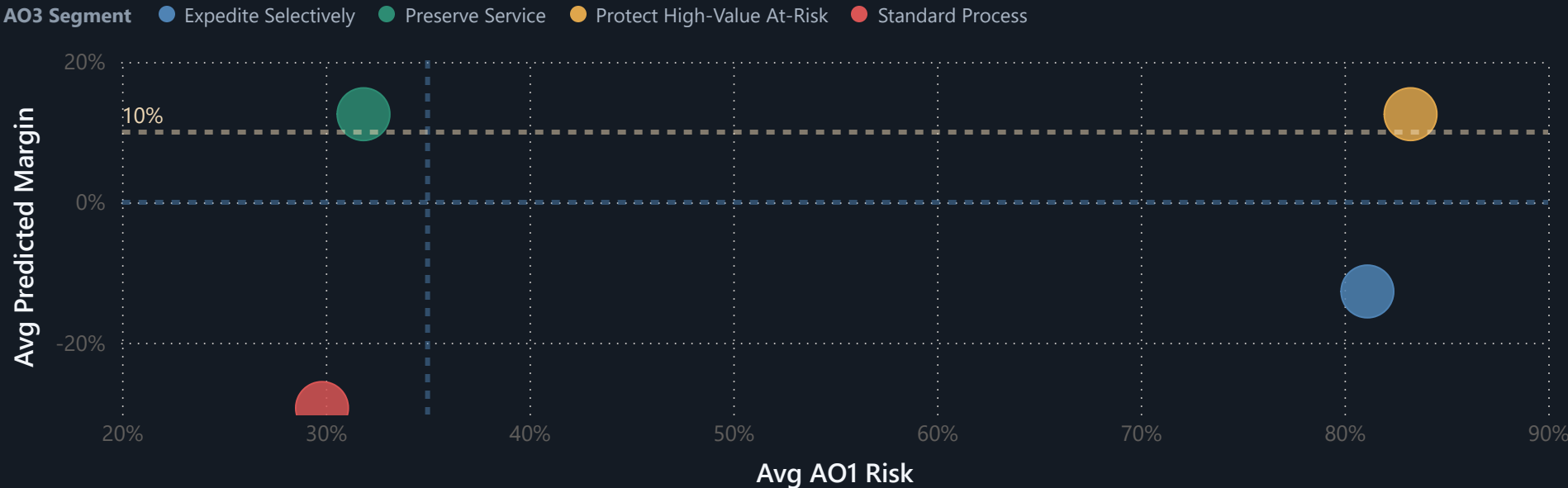
\$52.65

R²

0.012

Useful for triage, not for precise financial forecasting.

Risk x Margin Exposure by AO3 Segment



METHODOLOGY NOTE

Validation evidence: 28,883 rows · Operational exposure: 34,467 held-out scored items · Frozen upstream estimates

EXECUTIVE TAKEAWAY

112 order items require margin-protection review.

Frozen AO2 outputs estimate \$740,319 in aggregate profit with a 10.4% expected margin. Use the model for prioritization; validation evidence does not support precise item-level forecasting.

Where Does Margin Require Immediate Attention?

Protect expected profitability by identifying loss exposure, recent movement, and concentrated review priorities.

Profit Band

All

Order Date

22/04/2017

31/01/2018

Priority Segment

All

Risk Classification

All

Reset Filters

MODEL STATUS

Frozen AO2 outputs

GOVERNED DATA VERIFIED

EXPECTED PROFIT

\$740,319

Frozen AO2 portfolio estimate

EXPECTED MARGIN

10.4%

▲ +3.2% vs prior 4W

LATEST 4W PROFIT

\$22,089

▼ -31.8% vs prior 4W

NEGATIVE-PROFIT ITEMS

112

\$1,998 modeled exposure

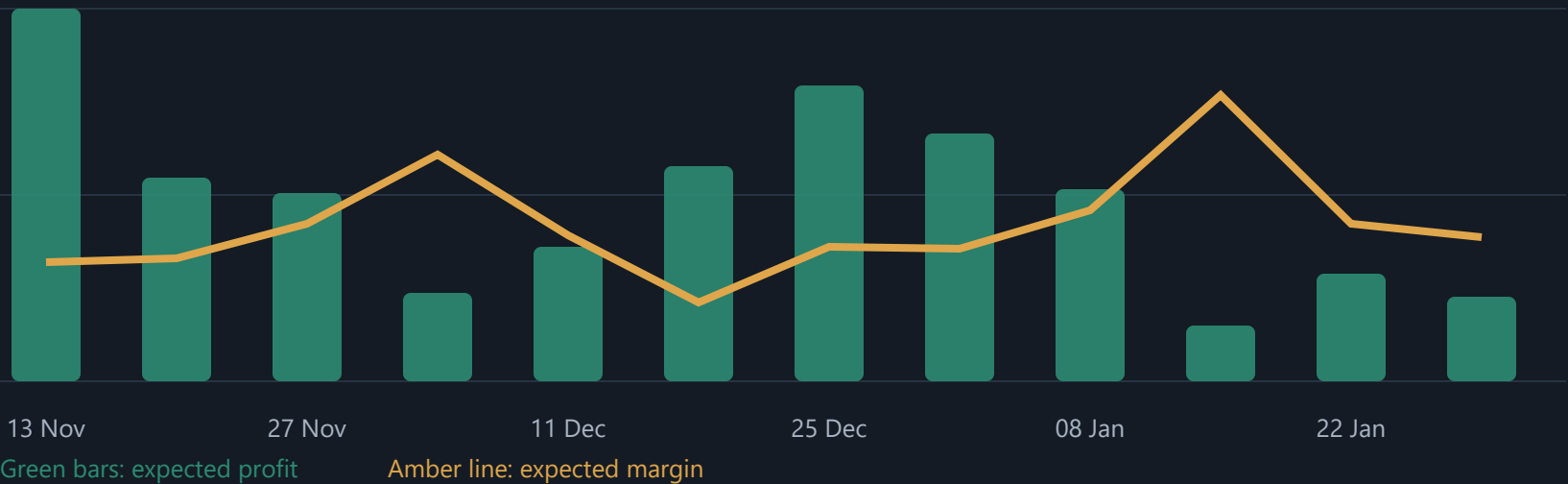
TOP 10 LOSS CONCENTRATION

57.3%

Focus first on concentrated loss exposure

WEEKLY EXPECTED PROFIT & MARGIN

Frozen profitability movement over the latest 12 available weeks.



MARGIN PROTECTION BRIEF

Separate volume movement from profitability efficiency before acting.

LATEST 4W PROFIT MOVEMENT

Expected profit decreased by 31.8%

Order value: -52.6% · Margin: +3.2%

LOSS EXPOSURE

\$1,998

112 negative-profit items

TOP 10 CONCENTRATION

57.3%

Review concentrated loss first

Profit fell while expected margin improved. Investigate volume composition and concentrated loss exposure before broad cost cutting.

MODEL EVIDENCE

RMSE ⓘ

\$95.62

MAE ⓘ

\$52.65

R² ⓘ

0.012

Useful for triage, not precise item-level forecasting. RMSE = overall error · MAE = average absolute error · R² = explained variance.

EXECUTIVE TAKEAWAY

Attack concentrated loss exposure first.

The ten largest predicted losses represent **57.3%** of total modeled loss exposure.

NEGATIVE-PROFIT REVIEW QUEUE

All modeled loss items ranked by expected loss exposure.

TOTAL LOSS QUEUE

112 items

ORDER ID	ITEM ID	ORDER VALUE	EXPECTED PROFIT	PRED. MARGIN	LATE RISK
66034	165038	\$328	-\$215.27	-65.6%	LOW
64292	160718	\$328	-\$137.69	-42.0%	HIGH
62660	156668	\$425	-\$127.56	-30.0%	LOW
64291	160713	\$83	-\$117.87	-142.0%	HIGH
75696	179011	\$83	-\$113.87	-137.7%	LOW

Frozen modeled outputs for prioritization. Product names require serving-layer enrichment.

Risk-Margin Operational Priorities

Focus preventive effort where delivery risk and expected value require action.

Profit Band

All

Order Date

22/04/2017

31/01/2018

Priority Segment

All

High Delivery Risk

All

Reset Filters

POLICY STATUS

Frozen AO3 matrix

GOVERNED DATA VERIFIED

SCORED ORDER ITEMS

34,467

PROTECT FIRST

13,752

PROTECT-FIRST SHARE

39.9%

VALUE AT RISK

\$2,816,571

SELECTIVE EXPEDITE

52

RISK-MARGIN PRIORITY MATRIX

Frozen AO3 segments convert model outputs into operating choices.

Reset Filters

HIGH RISK · LOW MARGIN

Expedite selectively

52

Review before premium-cost action.

HIGH RISK · HIGH MARGIN

Protect high value at risk

13,752

Prioritize pre-dispatch protection.

LOW RISK · LOW MARGIN

Standard process

60

Avoid special-cost interventions.

LOW RISK · HIGH MARGIN

Preserve service

20,603

Maintain quality without over-service.

PRIORITY ACTION QUEUE

01 · Protect first

13,752 items

Confirm dispatch readiness and escalate preventable delay risks.

02 · Expedite selectively

52 items

Intervene only when SLA or customer impact justifies cost.

03 · Preserve service

20,603 items

Maintain normal controls and avoid unnecessary escalation.

04 · Standard process

60 items

Use routine handling unless new operational alerts emerge.

METHODOLOGY NOTE

AO3 applies a deterministic 2x2 matrix to frozen AO1 risk and AO2 margin outputs. It does not retrain models, redefine thresholds, or claim realized intervention outcomes.



GOVERNANCE STATUS

AO1 RISK CUTOFF

35.0%

AO3 MARGIN CUTOFF

0.0%

DATA QA PASSED · 0 issue rows

Where Should Operations Allocate Attention First?

Balance preventive protection, selective expedite capacity, and stable-service preservation.

Profit Band

All

Priority Segment

All

Order Date

22/04/2017

31/01/2018

High Delivery Risk

All

Reset Filters

POLICY STATUS

Frozen AO3 matrix

GOVERNED DATA VERIFIED

ACTIVE REVIEW QUEUE

13,804

40.0% of scored items

PROTECT QUEUE

13,752

▲ +2.7% vs prior 4W

VALUE TO PROTECT

\$2,816,571

▼ -49.0% recent value movement

SELECTIVE EXPEDITE

52

\$8,564 queue value

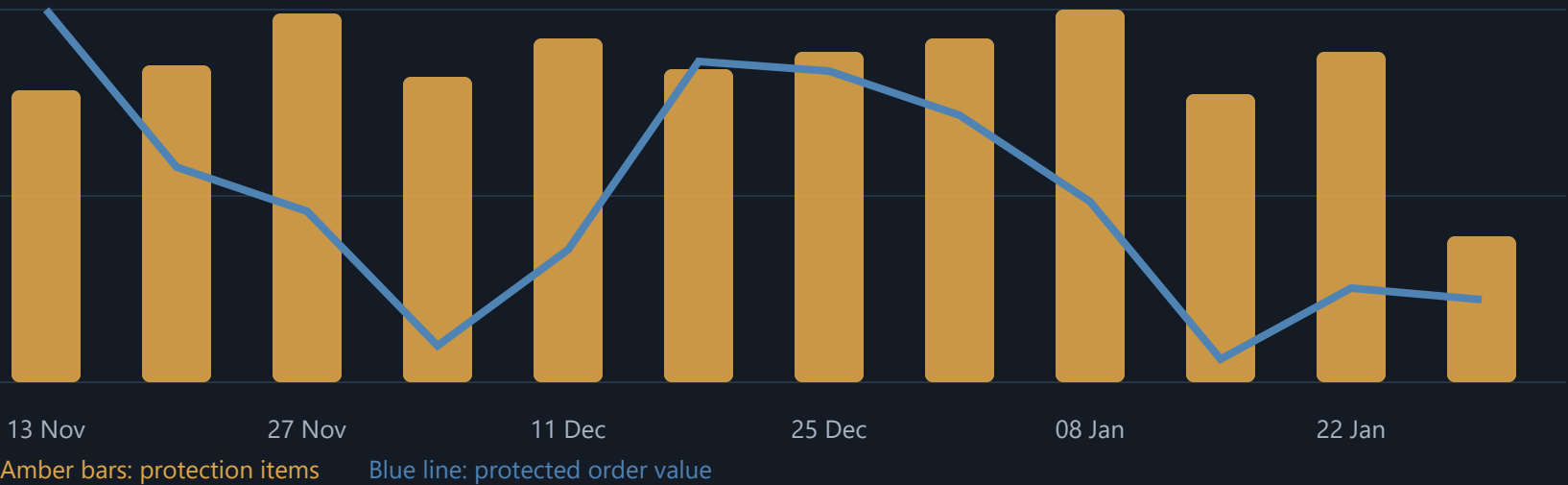
PRESERVE SERVICE

20,603

Avoid unnecessary intervention cost

WEEKLY PROTECTION CAPACITY

Protection workload and governed order value over the latest 12 available weeks.



OPERATIONAL ALLOCATION BRIEF

Apply different operating treatments to different AO3 queues.

PROTECTION CAPACITY

735 items in latest 4W

Queue movement +2.7% · Value movement -49.0%

SELECTIVE EXPEDITE

52 items

Review individually before spending

PRESERVE SERVICE

20,603 items

Avoid over-service costs

Scale preventive protection for the broad queue. Use selective expedite only after item-level review of financially fragile orders.

OPERATIONAL PERFORMANCE

Recent workload movement and intervention intensity.

PROTECTION QUEUE · LATEST 4W

735 items

+2.7% vs prior 4W

PROTECTED VALUE · LATEST 4W

\$90,290

-49.0% vs prior 4W

AVG VALUE PER PROTECTION ITEM

\$123

-50.3% vs prior 4W

PRESERVE SERVICE RATE

59.8%

Avoid unnecessary intervention cost

ATTENTION ALLOCATION

Use capacity according to operational treatment, not one blanket rule.

1 · SCALE PREVENTIVE PROTECTION

13,752 items · \$2,816,571

Broad queue. Confirm dispatch readiness early and manage exception capacity systematically.

3 · PRESERVE SERVICE WITHOUT OVER-SERVICE

20,603 items · \$4,316,203

Maintain standard quality controls and protect margin by avoiding unnecessary escalation.

2 · REVIEW EXPEDITE INDIVIDUALLY

52 items · \$8,564

Avoid automatic premium cost. Review SLA and customer impact before spending.

NEXT ANALYTICAL LENS

Customer · SLA · Shipping Mode · Route

Add a top-15 scatter after serving-layer enrichment to locate recurring operational hotspots.

SELECTIVE EXPEDITE

52 items

How Should Operations Adapt Capacity Now?

Track recent workload pressure, understand value intensity, and reserve premium intervention for the right exceptions.

Profit Band

All

Priority Segment

All

Order Date

22/04/2017

31/01/2018

High Delivery Risk

All

Reset Filters

POLICY STATUS

Frozen AO3 matrix

GOVERNED DATA VERIFIED

LATEST 4W PROTECTION QUEUE

735

+2.7% vs prior 4W

LATEST 4W PROTECTED VALUE

\$90,290

-49.0% vs prior 4W

AVG VALUE PER PROTECTION ITEM

\$123

-50.3% vs prior 4W

LATEST 4W SELECTIVE EXPEDITE

3

-62.5% vs prior 4W

1 · WHAT CHANGED? WEEKLY PROTECTION CAPACITY

Workload volume and protected order value over the latest 12 available weeks.



2 · WHY DOES IT MATTER?

Workload increased while value intensity declined.

The latest four-week protection queue grew **+2.7%**, while protected order value moved **-49.0%**. Average value per protection item fell **-50.3%**.

MANAGERIAL IMPLICATION

Do not apply premium handling uniformly. Scale low-cost preventive controls broadly and reserve expensive intervention for reviewed exceptions.

3 · WHAT SHOULD OPERATIONS DO?

Apply differentiated handling according to operational value and risk.

1 · SCALE PREVENTIVE PROTECTION

13,752 items

\$2,816,571 order value

Confirm dispatch readiness early. Apply scalable exception controls.

2 · REVIEW EXPEDITE INDIVIDUALLY

52 items

\$8,564 order value

Check SLA and customer impact before using premium capacity.

3 · PRESERVE SERVICE QUALITY

20,603 items

\$4,316,203 order value

Avoid over-service cost. Maintain normal quality controls.

Where Should Regional Attention Be Allocated First?

Separate high-volume exposure from disproportionate regional severity to focus operational review.

Profit Band

All

Priority Segment

All

Order Date

22/04/2017

31/01/2018

High Delivery Risk

All

Reset Filters

MAP COVERAGE

100.0%

GOVERNED LOCATIONS VERIFIED

COUNTRIES COVERED

66

12,652 mapped locations

HIGH-RISK ITEMS

13,804

40.0% of geographic portfolio

VALUE UNDER REVIEW

\$7,148,008

Governed order value

PROTECT-FIRST ITEMS

13,752

Regional preventive queue

SELECTIVE EXPEDITE

52

Review before premium spending

TOP VOLUME HOTSPOTS

Countries with the largest preventive-risk workload.

COUNTRY	HIGH-RISK WORKLOAD	ITEMS	RATE
Francia		2,337	40.4%
Alemania		1,705	40.4%
Reino Unido		1,169	37.0%
Mxico		917	39.4%
Italia		856	40.1%
Brasil		716	41.1%
Espaa		665	39.8%
Australia		528	39.0%
China		398	40.7%
Suecia		393	91.0%

DISPROPORTIONATE SEVERITY

Countries with the highest risk rate among portfolios of at least 100 items.

COUNTRY	RISK SEVERITY	ITEMS	RATE
Suecia		393	91.0%
Blgica		171	59.2%
Venezuela		95	56.9%
Per		49	45.4%
Singapur		46	44.2%
India		357	42.7%
Suiza		50	42.4%
Chile		44	41.9%
Guatemala		231	41.5%
Finlandia		50	41.3%

EXECUTIVE TAKEAWAY

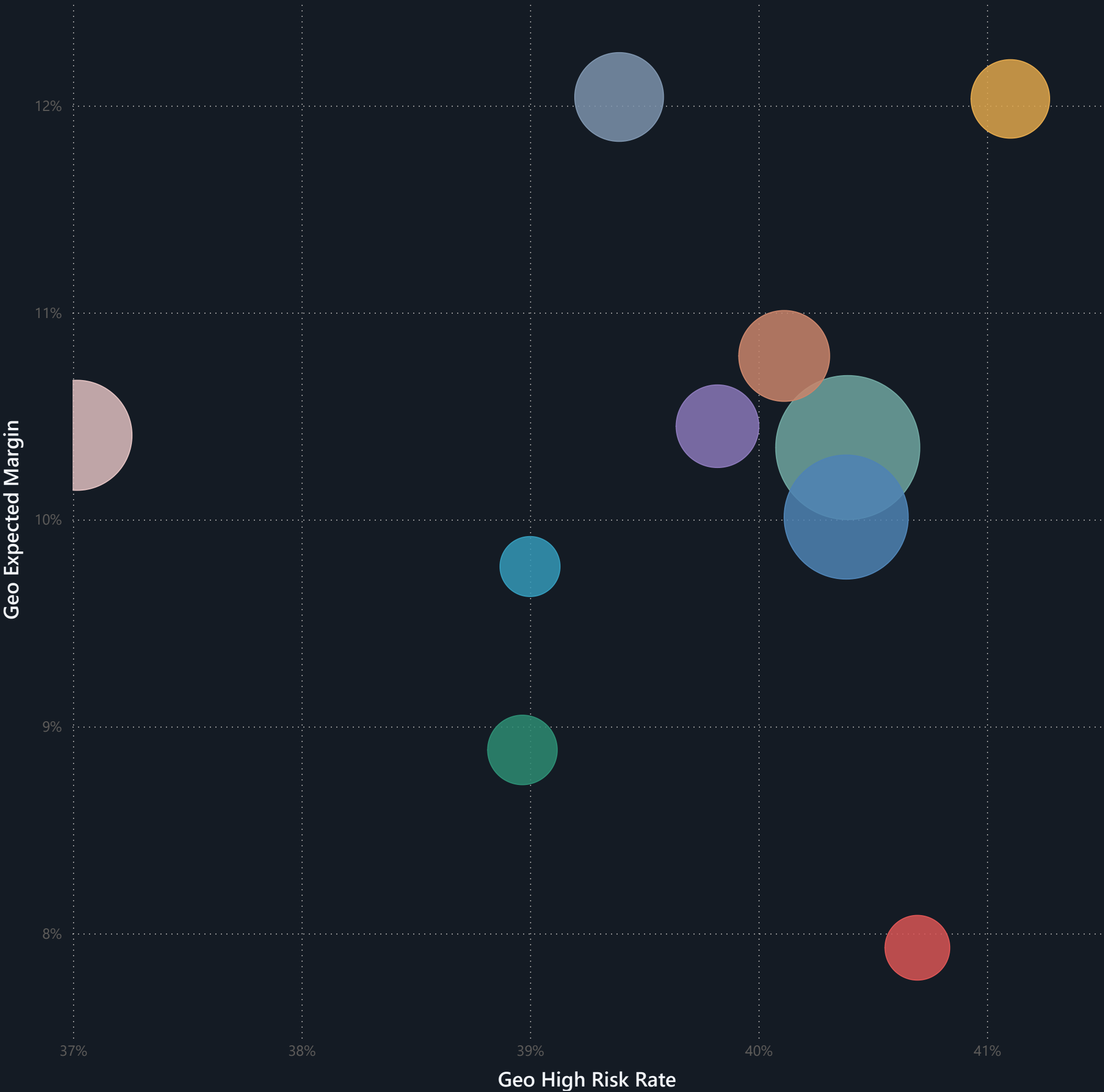
Separate workload scale from risk severity.

France and Germany lead the preventive queue by absolute volume. Sweden requires a different response: its high-risk rate reaches **91.0%**, indicating disproportionate severity despite lower volume.

Use the map and country scatter to distinguish capacity planning from root-cause investigation.

Geo High Risk Rate, Geo Expected Margin and Geo Order Value by map_location_country

map_location_country alemania australia brasil china espaa francia italia mxico pases_bajos reino_unido



Sum of high_risk_order_count by map_location_country

